

SUMMARY

Data Science and Engineering undergraduate specializing in full-stack development, scalable backend architecture, and secure system design for high-traffic platforms. Strong analytical problem solver recognized through competitive programming achievements (Codeforces Specialist, 750+ LeetCode solutions), highly dedicated to building efficient, user-centric software solutions that drive measurable business impact.

EDUCATION

National Institute of Technology (NIT), Jalandhar <i>B.Tech. in Data Science and Engineering Coursework: ML, NLP, Systems Design, Cloud</i>	2023 – Present CGPA: 8.29
Gobindgarh Public School, Mohali <i>Class 11–12 (Science Stream)</i>	2021 – 2023 89%

SKILLS

AI, ML & Data Science: Deep Learning (CNN/RNN), NLP, LLM, TensorFlow, Scikit-learn, OpenCV, Pandas, LangChain, LangGraph, Agentic AI, RAG Systems, Vector Databases
Programming Languages: Python, Golang, C++, JavaScript
Web & Full-Stack: React.js, Next.js, Node.js, Express.js, Tailwind CSS, WebSockets, tRPC, gRPC, Prisma ORM
Cloud, Databases & Tools: PostgreSQL, MongoDB, MySQL, Docker, Kubernetes, AWS, Linux, Git, REST APIs
Core Competencies: Data Structures, Object-Oriented Programming (OOP), DBMS, System Design, Operating Systems

EXPERIENCE

- Advance Application Engineering (AEH) Intern — AI Agent Development — Accenture, Bangalore — Jun 2026–Jul 2026:**
 - Architected enterprise **AI agents** using **RAG** pipelines to ground LLMs in multi-tenant data, eliminating context drift.
 - Engineered semantic search infrastructure using **vector embeddings**, boosting context retrieval precision by **40%**.
 - Automated document ingestion with **Python** and built **React.js** dashboards to track real-time agent token metrics and latency.
- CTP (NITJ Placement Portal) [Live Site] — Backend Lead — NIT Jalandhar — 2024–2026:**
 - Built high-throughput **Node.js/Express.js** backend serving **1,200+ users** across **25+ REST APIs** for recruitment workflows.
 - Restructured **MongoDB** indexing strategies, cutting data retrieval time by **35%** under **10k+ daily requests**.
 - Implemented granular **RBAC** security protocols and **React.js** interfaces, accelerating shortlisting cycles by **20%**.
- Third Space Travel (Travel Booking Platform) [Live Site] — Full-Stack Developer — On-Site — 2026:**
 - Built a full-stack platform using **React**, **TypeScript**, and **Prisma/MySQL**, optimizing queries to reduce latency by **50ms**.
 - Integrated **PhonePe** payment SDK and **tRPC** endpoints, achieving a **99.9% success rate** and lowering drop-offs by **20%**.
 - Developed modular **SCSS** and **Framer Motion** components, lifting mobile retention by **25%** with a **95+ Lighthouse** score.

PROJECTS

- Shadow-Room: E2EE Secure Chat System — 2026** [Live Demo](#) | [Repo](#):
 - Built an end-to-end encrypted (E2EE) messaging platform utilizing **Golang**, **React.js**, and **WebSockets** for low-latency, real-time asynchronous communication across distributed networks.
 - Implemented cryptographic handshakes (**RSA**, **MD5**, **SHA-256**) and the **Double Ratchet Algorithm** to establish forward secrecy, message integrity, and group scaling via Sender Keys.
 - Integrated multi-factor authentication and executed **Protocol Buffers** modifications to support high-throughput concurrent user sessions while ensuring zero data leaks.
- Secure Token Allocation System — NIT Jalandhar — 2026** [Live Demo](#) | [Repo](#):
 - Developed a cryptographically secure token generation and tracking system to automate physical reporting for **1,200+ incoming students**.
 - Implemented **AES-256** encryption and secure hashing to prevent token forgery and ensure a tamper-proof queue management process.
 - Built a responsive **React.js** dashboard to monitor live allocations, reducing manual verification bottlenecks and wait times by **60%**.
- DataPulse: Operations Management Hub — 2025** [Live Demo](#) | [Repo](#):
 - Engineered a scalable data operations API utilizing **Flask** and **GraphQL** to aggregate and serve complex datasets dynamically.
 - Designed custom **GraphQL** resolvers to eliminate data over-fetching, reducing frontend API payload sizes by **70%**.
 - Integrated **Redis** caching with a **PostgreSQL** database to process high-frequency queries, maintaining a **200ms** response time.

ACHIEVEMENTS & LEADERSHIP

- Codeforces & LeetCode:** Achieved **Specialist** rank on Codeforces (Peak Rating: **1502**, Global Rank **676** in Div. 2); ranked in the **Top 5%** globally on LeetCode with **750+** problems solved (**1,050+** across all platforms).
- AWS Club (NIT Jalandhar) — Technical Lead:** Spearheaded the campus cloud community by organizing hands-on bootcamps on **Cloud Architecture**, **DevOps**, and **GenAI**, mentoring **300+** students in scalable backend systems and containerization.
- Speed Quant (UTKANSH'25) — First Place:** Secured **1st Rank** out of **200+** participants in the flagship quantitative competition at NIT Jalandhar, solving high-pressure mathematical and algorithmic problems.